

- A value of "bison" would be acceptable as the string size of 5 is the lower end of the range of the constraint.
- A value of "beaver" would be acceptable as the string size of 6 is between the lower and upper range of the constraint.
- A value of "buffalo" would be acceptable as the string size of 7 is the upper end of the range of the constraint.
- A value of "bontebok" would violate the constraints as the size of 8 is more than the upper end of the constraint range, which is 7.

7.18.3.6. Effective Maximum for Character String Data Type

A server SHALL support up to the maximum in *char_constraint* for a character string data type. The character string data SHALL NOT contain more than z Unicode codepoints.

Example: A string with a constraint of "max 128{32}" dictates that the server provide for a 128 byte string, but the string may contain up to 32 Unicode codepoints.

7.18.3.7. Nullable in Range

If data is *nullable* then NULL SHALL be a valid value.

If the data type is a list or derived from a list, and the list is nullable, then a length of 0 (zero) SHALL be supported, and defined in the constraint column.

If the data type is an octet string, or derived from an octet string (e.g. character string), and the data is nullable, then a length of 0 (zero) SHALL be supported, and defined in the constraint column.

7.18.3.8. Enumeration Constraint

When the constraint for an element is 'all', if any attempt is made in an interaction to use a value that is not defined in the data type definition associated with the element, the result for the interaction SHALL be a constraint violation with the response defined by the actual interaction, e.g. an attribute write or a cluster command.

NOTE

Values that are well-defined could still result in a constraint error or other violation due to features unsupported by a server, the current system state, or other reasons associated with the particular device.

7.18.4. Fallback Column

A fallback value defined in this column is not meant to be the value used when the server returns to factory default settings. Specified conformance for data fields may be optional or change over time. A fallback value is defined to complete dependencies when the actual data field value is not present.

A data field SHALL have a defined fallback value when: